



The University of Danang

University of Science and Technology

MATHEMATICS FOR COMPUTER SCIENCE

Chap 5. Probability



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References

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Probability

- **Calculation**
- Events
- Random variables
- Markov chains

Calculation

- **A probability**

- is a real number between 0 and 1 that indicates how probable an event is.
- If an event is certain to happen, its probability is 1, and if an event is impossible, its probability is 0.
- The probability of an event is denoted $P(\dots)$ where the three dots describe the event.

Calculation

- **A probability**

- For example,
 - when throwing a dice, the outcome is an integer between 1 and 6, and the probability of each outcome is $1/6$.
 - we can calculate the following probabilities:
 - $P(\text{the outcome is 4}) = 1/6$
 - $P(\text{the outcome is not 6}) = 5/6$
 - $P(\text{the outcome is even}) = 1/2$

Calculation

- To calculate the probability of an event, we can either use combinatorics or simulate the process that generates the event.
 - An example: Let us calculate the probability of drawing three cards with the same value from a shuffled deck of cards (for example, 9 ♠, 9 ♣, 9 ♥).

Calculation

- To calculate the probability of an event, we can either use combinatorics or simulate the process that generates the event.

- An example:

- **Method 1:** We can calculate the probability using the formula

$$\frac{\text{number of desired outcomes}}{\text{total number of outcomes}}$$

- In this problem, the desired outcomes are those in which the value of each card is the same.
 - There are $13C_4^3$ such outcomes, because there are 13 possibilities for the value of the cards and C_4^3 ways to choose 3 suits from 4 possible suits.

There are a total of C_{52}^3 outcomes, because we choose 3 cards from 52 cards. Thus, the probability of the event is $\frac{13C_4^3}{C_{52}^3} = \frac{13 \times 4}{22100} = \frac{1}{425}$

Calculation

- An example:
 - **Method 2:** Another way to calculate the probability is to simulate the process that generates the event.
 - We draw three cards, so the process consists of three steps. We require that each step of the process is successful.
 - Drawing the first card certainly succeeds, because there are no restrictions
 - The second step succeeds with probability $\frac{3}{51}$, because there are 51 cards left and 3 of them have the same value as the first card.
 - The third step succeeds with probability $\frac{2}{50}$
 - The probability that the entire process succeeds is $1 \times \frac{3}{51} \times \frac{2}{50} = \frac{1}{425}$

Probability

- Calculation
- **Events**
- Random variables
- Markov chains
- Randomized algorithms

Events

- An event in probability theory can be represented as a set

$$A \subset X$$

where X contains all possible outcomes and A is a subset of outcomes.

- For example, when drawing a dice, the outcomes are

$$X = \{1, 2, 3, 4, 5, 6\}$$

- Now, for example, the event "**the outcome is even**" corresponds to the set

$$A = \{2, 4, 6\}$$

Events

- Each outcome x is assigned a probability $p(x)$.
 - Then, the probability $P(A)$ of an event A can be calculated as a sum of probabilities of outcomes using the formula

$$P(A) = \sum_{x \in A} p(x)$$

- For example, when throwing a dice, $p(x) = \frac{1}{6}$ for each outcome x , so the probability of the event "**the outcome is even**" is

$$p(2) + p(4) + p(6) = \frac{1}{2}$$

- The total probability of the outcomes in X must be 1, i.e., $P(X) = 1$

Events

- Since the events in probability theory are sets, we can manipulate them using standard set operations:
 - The complement $\bar{A}$ means "**A does not happen**". For example, when throwing a dice, the complement of $A = \{2,4,6\}$ is $\bar{A} = \{1,3,5\}$
 - The union $A \cup B$ means "**A or B happen**". For example, the union of $A = \{2,5\}$ and $B = \{4,5,6\}$ is $A \cup B = \{2,4,5,6\}$
 - The intersection $A \cap B$ means "**A and B happen**". For example, the intersection of $A = \{2,5\}$ and $B = \{4,5,6\}$ is $A \cap B = \{5\}$

Events

- Complement
 - The probability of the complement $\bar{A}$ is calculated using the formula
$$P(\bar{A}) = 1 - P(A)$$
 - Sometimes, we can solve a problem easily using complements by solving the opposite problem.
 - For example, the probability of getting at least one six when throwing a dice ten times is $1 - \left(\frac{5}{6}\right)^{10}$
 - Here $\frac{5}{6}$ is the probability that the outcome of a single throw is not six, and $\left(\frac{5}{6}\right)^{10}$ is the probability that none of the ten throws is a six. The complement of this is the answer to the problem.

Events

- Union

- The probability of the union $A \cup B$ is calculated using the formula

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- For example, when throwing a dice, the union of the events

$A = \text{"the outcome is even"}$ and $B = \text{"the outcome is less than 4"}$:

$A \cup B = \text{"the outcome is even or less than 4"}$

- and its probability is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{2} + \frac{1}{2} - \frac{1}{6}$$

Events

- Union
 - If the events A and B are **disjoint**, i.e., $A \cap B$ is empty, the probability of the event $A \cup B$ is simply

$$P(A \cup B) = P(A) + P(B)$$

Events

- Conditional probability
 - The conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- the probability of A assuming that B happens.
- when calculating the probability of A , we only consider the outcomes that also belong to B

Events

- Conditional probability
 - The conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- Using the previous sets: $P(A|B) = \frac{1}{3}$
 - the outcomes of B are $\{1,2,3\}$, and one of them is **even**
 - this is the probability of an even outcome if we know that the outcome is between 1...3

Events

- Intersection

- Using conditional probability, the probability of the intersection $A \cap B$ can be calculated using the formula

$$P(A \cap B) = P(A)P(B|A)$$

- Events A and B are independent if

$$P(A|B) = P(A) \text{ and } P(B|A) = P(B)$$

Events

- Intersection
 - For example, when drawing a card from a deck
 - The events $A = \text{"the suit is clubs"}$ and $B = \text{"the value is four"}$ are independent
 - Hence the event $A \cap B = \text{"the card is the four of clubs"}$ happens with probability

$$P(A \cap B) = P(A)P(B) = \frac{1}{4} \times \frac{1}{13} = \frac{1}{52}$$

Probability

- Calculation
- Events
- **Random variables**
- Markov chains

Random variables

- A random variable is a value that is generated by a random process
 - When throwing two dice, a possible random variable is
$$X = \text{"the sum of the outcomes"}$$
 - If the outcomes are $[4,6]$ (meaning that we first throw a four and then a six), then the value of X is 10
 - We denote $P(X = x)$ the probability that the value of a random variable X is x
 - When throwing two dice, $P(X = 10) = 3/36$, because the total number of outcomes is 36 and there are three possible ways to obtain the sum 10:
$$[4,6], [5,5] \text{ and } [6,4]$$

Random variables

- Expected value

- The expected value $E[X]$ indicates the average value of a random variable X
- The expected value can be calculated as the sum

$$\sum_x P(X = x)x$$

where x goes through all possible values of X

- For example, when throwing a dice, the expected outcome is:

$$\frac{1}{6} \times 1 + \frac{1}{6} \times 2 + \frac{1}{6} \times 3 + \frac{1}{6} \times 4 + \frac{1}{6} \times 5 + \frac{1}{6} \times 6 = \frac{7}{2}$$

Random variables

- Expected value
 - A useful property of expected values is **linearity**. It means that the sum $E[X_1 + X_2 + \dots + X_n]$ always equals the sum $E[X_1] + E[X_2] + \dots + E[X_n]$
 - This formula holds even if random variables depend on each other
 - For example, when throwing two dice, the expected sum is

$$E[X_1 + X_2] = E[X_1] + E[X_2] = \frac{7}{2} + \frac{7}{2} = 7$$

Random variables

- Expected value
 - Let us now consider a problem where n balls are randomly placed in n boxes, and our task is to calculate the expected number of empty boxes
 - Each ball has an equal probability to be placed in any of the boxes. For example, if $n = 2$, the possibilities are as follows:



Random variables

- Expected value
 - Let us now consider a problem where n balls are randomly placed in n boxes, and our task is to calculate the expected number of empty boxes
 - Each ball has an equal probability to be placed in any of the boxes. For example, if $n = 2$, the possibilities are as follows:



- The expected number of empty boxes is

$$\frac{0 + 0 + 1 + 1}{4} = \frac{1}{2}$$

Random variables

- Expected value
 - Each ball has an equal probability to be placed in any of the boxes. For example, if $n = 2$, the possibilities are as follows:
 - In the general case, the probability that a single box is empty is

$$\left(\frac{n-1}{n}\right)^n$$

because no ball should be placed in it

- Using linearity, the expected number of empty boxes is

$$n \times \left(\frac{n-1}{n}\right)^n$$

Random variables

- Distributions
 - The distribution of a random variable X shows the probability of each value that X may have.
 - The distribution consists of values $P(X = x)$
 - For example, when throwing two dice, the distribution for their sum is:

x	2	3	4	5	6	7	8	9	10	11	12
$P(X = x)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

Random variables

- Normal distribution

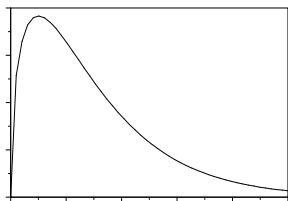
- A key parametric statistical model is the standard normal distribution $N(0; 1)$ whose cumulative distribution function

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-y^2/2} dy$$

- The importance of the normal model is due to the Central Limit Theorem (CLT) as well as the remarkable analytical properties of the normal density function.
- The CLT states that the sum of many independent and relatively small random contributions is approximately normally distributed.

Random variables

- Normal distribution: CENTRAL LIMIT THEOREM
 - If a random sample is drawn from any population, the sampling distribution of the sample mean is approximately normal for a sufficiently large sample size
 - The larger the sample size, the more closely the sampling distribution will resemble a normal distribution.

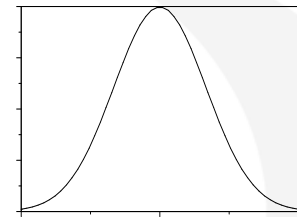


X

Random Variable
 (Population) Distribution

Random Sample
 $(X_1, X_2, X_3, \dots, X_n)$

as $n \rightarrow \infty$



$\bar{X}$

Sample Mean
 Distribution

Random variables

- Normal distribution
 - We write $N(0; \sigma)$ if the random variable has a normal distribution with mean zero and standard deviation σ
 - We view of the CLT it is natural to take as a model of the random noise since the noise is an accumulation of all minor independent factors neither of which having a dominating effect of the response variable.
 - Notice that the standard deviation plays the role of a scale parameter of the normal distribution in that $\frac{\epsilon}{\sigma} N(0; 1)$.

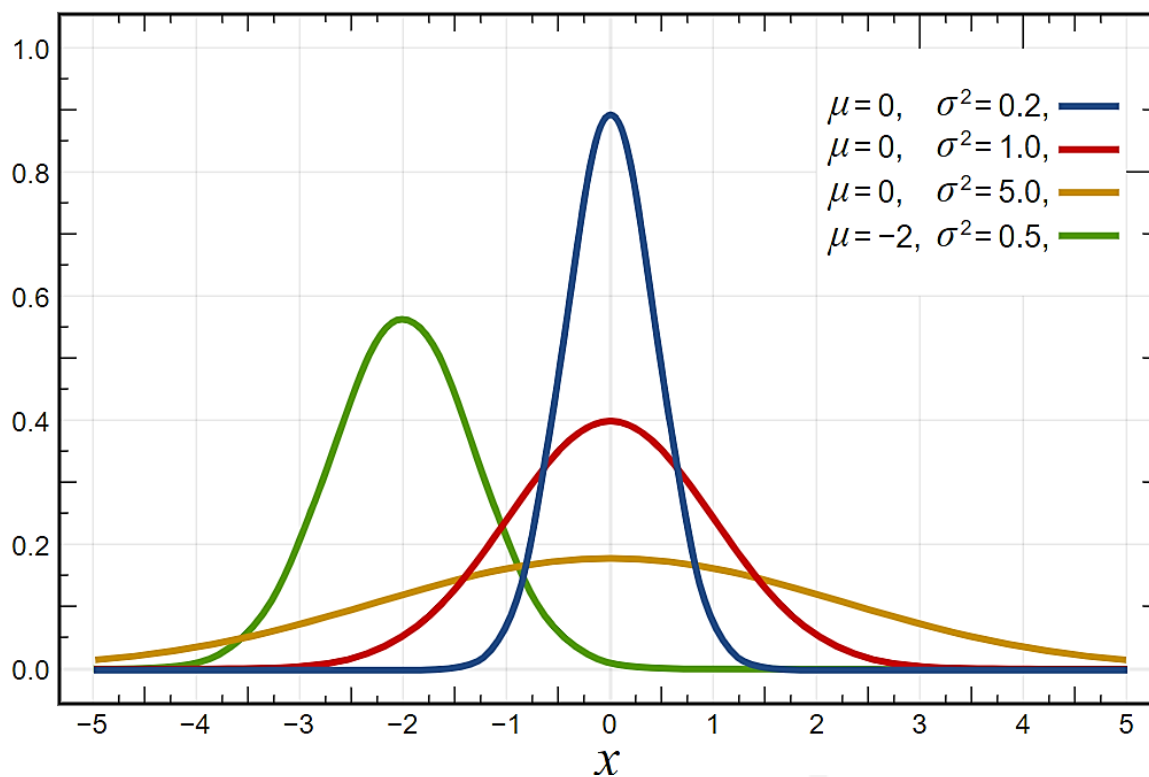
Random variables

- Normal distribution
 - Notice that the standard deviation plays the role of a scale parameter of the normal distribution in that $\frac{\epsilon}{\sigma} N(0; 1)$
 - A random response variable Y having a normal distribution $N(\mu; \sigma)$ with mean μ and the standard deviation σ can be viewed as the sum of a constant signal μ and a noise ϵ

$$Y = \mu + \epsilon, \quad \epsilon \sim N(0, \sigma)$$

Random variables

- Normal distribution



Random variables

- Mixture of normal distributions
 - Suppose we are given k normally distributed random variables

$$X_1 \sim N(\mu_1, \sigma_1), \dots, X_k \sim N(\mu_k, \sigma_k)$$

- Define $Y = X_i$ using a random index i taking one of the values $1, \dots, k$ with probabilities $w_1, \dots, w_k$, so that

$$w_1 + \dots + w_k = 1.$$

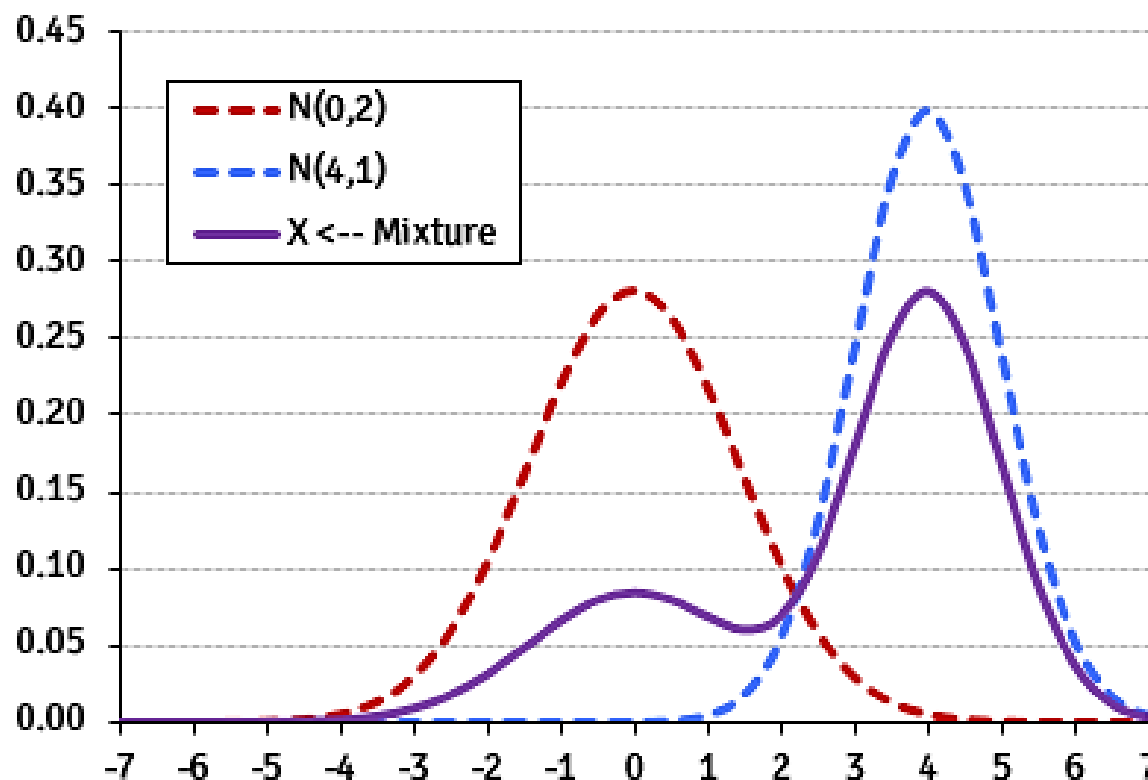
Random variables

- Mixture of normal distributions
 - We have the following expressions for the mean $\mu = E(Y)$ and variance $\sigma^2 = \text{Var}(Y)$

$$\mu = w_1\mu_1 + \dots + w_k\mu_k,$$
$$\sigma^2 = \sum_{j=1}^k w_j(\mu_j - \mu)^2 + \sum_{j=1}^k w_j\sigma_j^2.$$

Random variables

- Mixture of normal distributions



Random variables

- One-way layout model
 - Suppose the expectation of the response variable is a function of a single main factor having I different levels so that

$$Y(i) = \mu_i + \epsilon, \quad \epsilon \sim N(0, \sigma), \quad i = 1, \dots, I.$$

- Sometimes it is useful to use a representation of the overall mean

$$\mu_i = \mu + \alpha_i,$$

$$\mu = \frac{\mu_1 + \dots + \mu_I}{I}$$

- The effect of the main factor at the level i

$$\alpha_i = \mu_i - \mu$$

$$\sum_{i=1}^I \alpha_i = 0.$$

Random variables

- Two-way layout model
 - In the case of two main (categorical) factors

$$Y(i, j) = \mu_{ij} + \epsilon, \quad \epsilon \sim N(0, \sigma), \quad i = 1, \dots, I, \quad j = 1, \dots, J, \quad \mu_{ij} = \mu + \alpha_i + \beta_j + \delta_{ij},$$

- where μ – overall mean and α, β – two main effects at their respective levels,

so that

$$\sum_{i=1}^I \alpha_i = \sum_{j=1}^J \beta_j = 0.$$

$$\mu = \frac{1}{IJ} \sum_{i=1}^I \sum_{j=1}^J \mu_{ij},$$

$$\alpha_i = \frac{1}{J} \sum_{j=1}^J (\mu_{ij} - \mu),$$

$$\beta_j = \frac{1}{I} \sum_{i=1}^I (\mu_{ij} - \mu)$$

Random variables

- Two-way layout model

- The term δ_{ji} defined by $\delta_{ij} = \mu_{ij} - \mu - \alpha_i - \beta_j$

describes the interaction between the two main factors.

- Notice that $\sum_{i=1}^I \delta_{ij} = 0, \quad j = 1, \dots, J, \quad \sum_{j=1}^J \delta_{ij} = 0, \quad i = 1, \dots, I.$

- In the simple case with all $\delta_{ji} = 0$, there is no interaction and the main factors contribute additively:

$$\mu_{ij} = \mu + \alpha_i + \beta_j.$$

- In general, at different combinations of levels the two factors may interact either negatively or positively

Random variables

- Multiple regression model

- A model based on a linear relationship between $(p - 1)$ predictors and the response variable

$$Y(x_1, \dots, x_{p-1}) = \beta_0 + \beta_1 x_1 + \dots + \beta_{p-1} x_{p-1} + \epsilon, \quad \epsilon \sim N(0, \sigma).$$

- This is a flexible setting, it for example covers the one-way layout model with $I = p$ as a special case.

- Indeed, in the framework of the one-way layout model we can define parameters

$(\beta_0, \dots, \beta_{p-1})$ by

$$\mu_1 = \beta_0,$$

$$\mu_2 = \beta_0 + \beta_1,$$

$$\mu_3 = \beta_0 + \beta_2,$$

...

$$\mu_p = \beta_0 + \beta_{p-1},$$

Random variables

- Multiple regression model
 - Introduce dummy variables $(x_1, \dots, x_{p-1})$ taking values 0 and 1
 - It is easy to see that setting the levels of the main factor of the one-way layout model as

$$\text{level 1 : } (x_1, \dots, x_{p-1}) = (0, 0, 0, \dots, 0, 0)$$

$$\text{level 2 : } (x_1, \dots, x_{p-1}) = (1, 0, 0, \dots, 0, 0)$$

$$\text{level 3 : } (x_1, \dots, x_{p-1}) = (0, 1, 0, \dots, 0, 0)$$

...

$$\text{level } p : (x_1, \dots, x_{p-1}) = (0, 0, 0, \dots, 0, 1)$$

⇒ We arrive at a particular example of the multiple regression model

Random variables

- Binomial distribution

- Binomial distribution $X \sim \text{Bin}(n, p)$:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad 0 \leq k \leq n, \quad E(X) = np, \quad \text{Var}(X) = np(1 - p)$$

$$\binom{n}{k} = \frac{n!}{k!(n - k)!}$$

- A binomially distributed variable X is the sum of n independent random variables each having a Bernoulli distribution $\text{Bin}(1, p)$
- This leads to the important example of the CLT giving the normal approximation for the binomial distribution.

Random variables

- Binomial distribution
 - With a continuity correction the claim is that for $X \sim \text{Bin}(n, p)$

$$P(X \leq k - 1) = P(X < k) \approx \Phi\left(\frac{k - \frac{1}{2} - np}{\sqrt{np(1-p)}}\right)$$

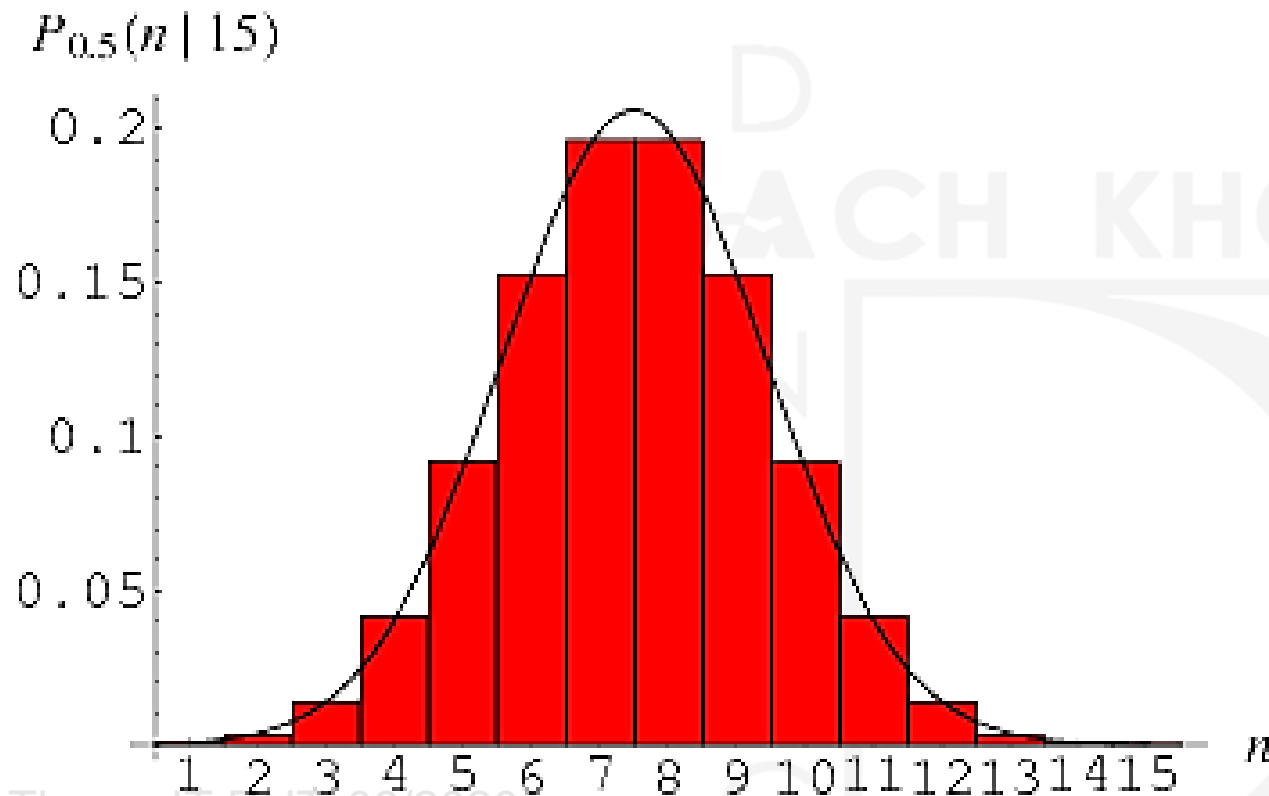
- The rule of thumb says

$\text{Bin}(n, p) \approx N(np, \sqrt{np(1-p)})$, if both $np \geq 5$ and $nq \geq 5$.

$$E[X] = np.$$

Random variables

- Binomial distribution



Random variables

- The Bernoulli distribution

- is a special case of the binomial distribution where a single trial is conducted (so n would be 1 for such a binomial distribution)
- It is also a special case of the two-point distribution, for which the possible outcomes need not be 0 and 1

$$P(n) = \begin{cases} 1 - p & \text{for } n = 0 \\ p & \text{for } n = 1, \end{cases}$$

which can also be written

$$P(n) = p^n (1 - p)^{1-n}.$$

The corresponding distribution function is

$$D(n) = \begin{cases} 1 - p & \text{for } n = 0 \\ 1 & \text{for } n = 1. \end{cases}$$

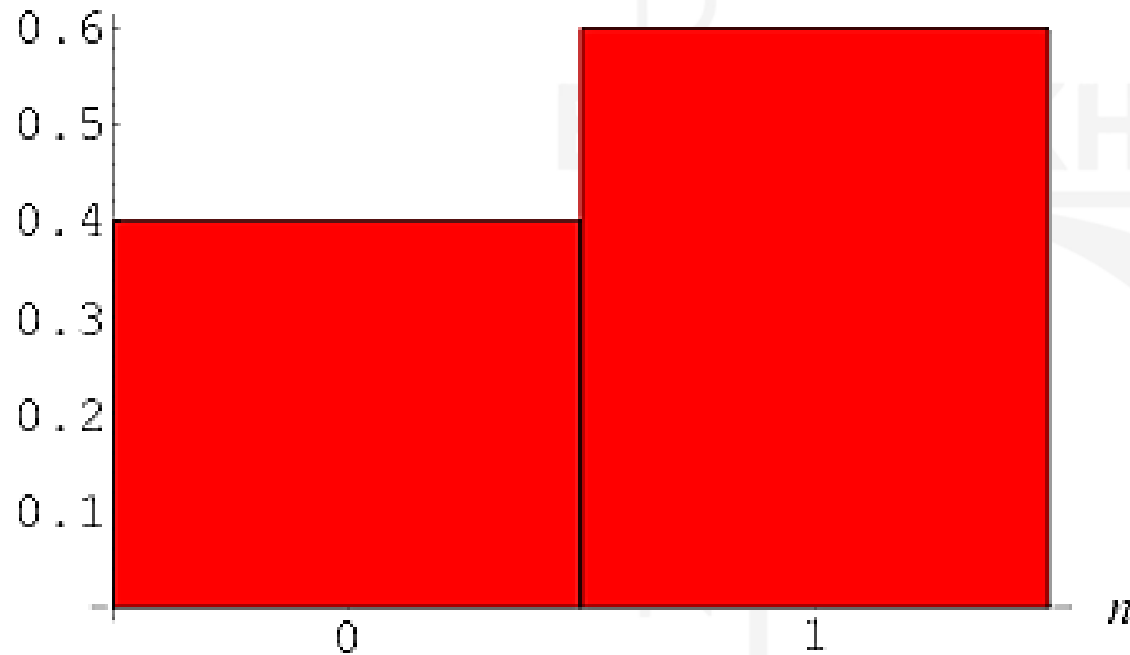
The expected value of a Bernoulli random variable X

$$E(X) = p$$

Random variables

- The Bernoulli distribution

$P(n)$ for $p = 0.6$



Random variables

- Uniform distribution
 - The random variable X has n possible values $a, a + 1, \dots, b$ and the probability of each value is $\frac{1}{n}$
 - For example, when throwing a dice, $a = 1, b = 6$ and for each value x .
$$P(X = x) = \frac{1}{6}$$
 - The expected value of X in a uniform distribution is

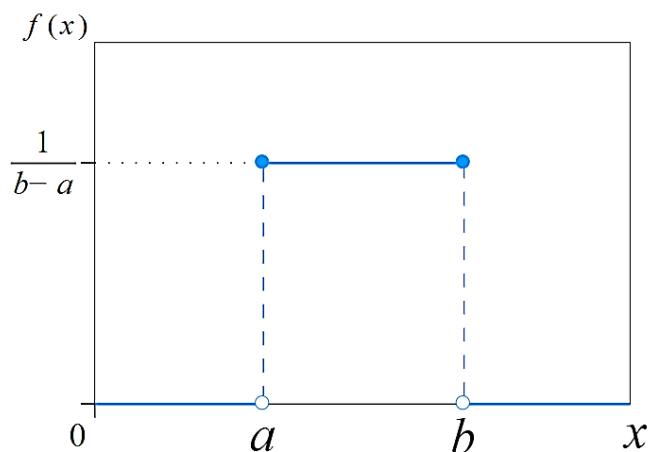
$$E[X] = \frac{a + b}{2}$$

- A uniform distribution, sometimes also known as a rectangular distribution, is a distribution that has constant probability

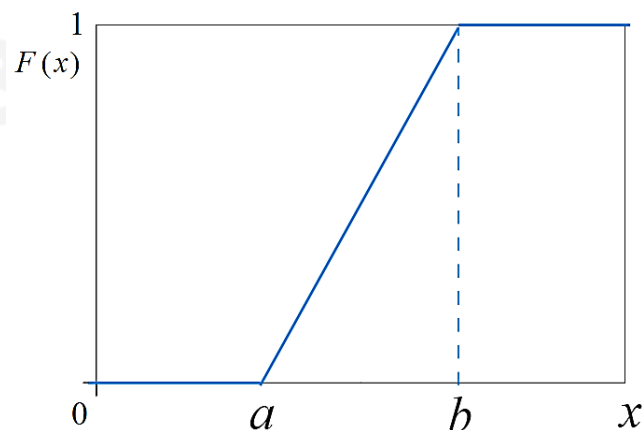
Random variables

- Uniform distribution

- The probability density function and cumulative distribution function for a continuous uniform distribution on the interval $[a, b]$



$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b, \\ 0 & \text{for } x < a \text{ or } x > b \end{cases}$$



$$F(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } a \leq x \leq b \\ 1 & \text{for } x > b \end{cases}$$

Random variables

- Uniform distribution
 - In terms of mean μ and variance σ^2 , the probability density may be written as:

$$f(x) = \begin{cases} \frac{1}{2\sigma\sqrt{3}} & \text{for } -\sigma\sqrt{3} \leq x - \mu \leq \sigma\sqrt{3} \\ 0 & \text{otherwise} \end{cases}$$

- In mean and variance notation, the cumulative distribution function is:

$$F(x) = \begin{cases} 0 & \text{for } x - \mu < -\sigma\sqrt{3} \\ \frac{1}{2} \left(\frac{x - \mu}{\sigma\sqrt{3}} + 1 \right) & \text{for } -\sigma\sqrt{3} \leq x - \mu < \sigma\sqrt{3} \\ 1 & \text{for } x - \mu \geq \sigma\sqrt{3} \end{cases}$$

Random variables

- Geometric distribution

- In a geometric distribution, the probability that an attempt succeeds is p , and we continue until the first success happens.
- The random variable X counts the number of attempts needed, and the probability of a value x is

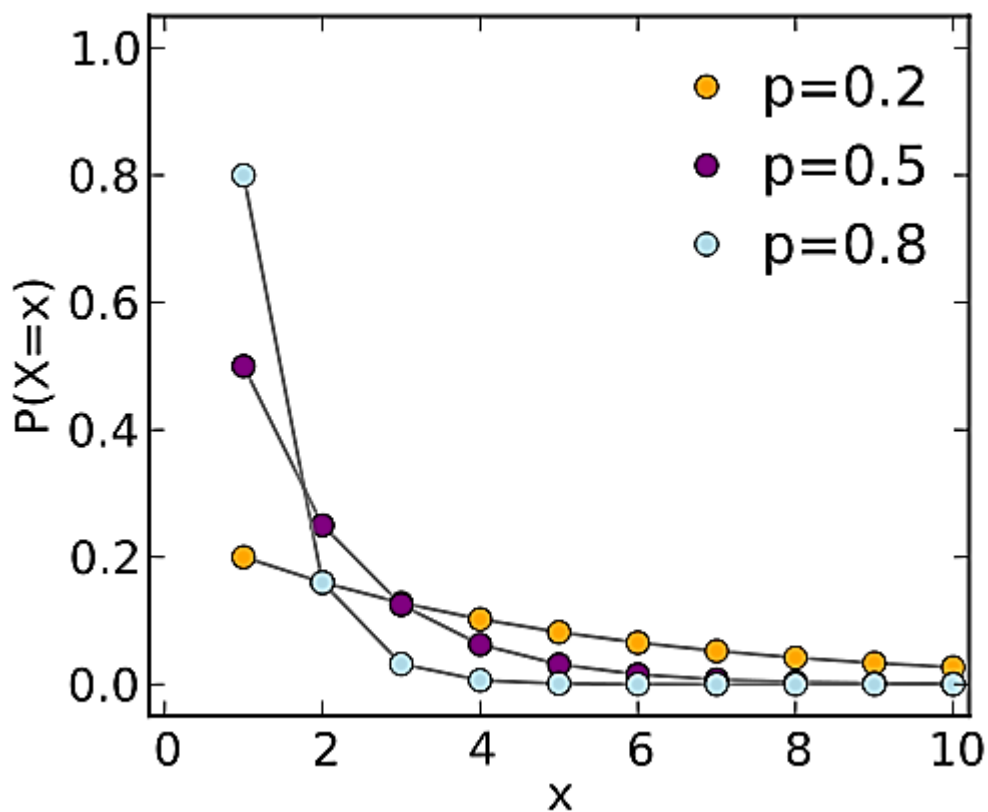
$$P(X = x) = (1 - p)^{x-1} p,$$

where $(1 - p)^{x-1}$ corresponds to the unsuccessful attempts and p corresponds to the first successful attempt

- For example, if we throw a dice until we throw a six, the probability that the number of throws is exactly 4 is $\left(\frac{5}{6}\right)^3 \frac{1}{6}$
- The expected value of X in a geometric distribution is $E[X] = \frac{1}{p}$.

Random variables

- Geometric distribution



Random variables

- Hypergeometric distribution

- Hypergeometric distribution $X \sim Hg(N; n; p)$ describes the number X of black balls among n balls drawn without replacement from a box with N balls of which N_p balls are black:

$$P(X = k) = \frac{\binom{Np}{k} \binom{N(1-p)}{n-k}}{\binom{N}{n}}, \quad \max(0, n - N(1 - p)) \leq k \leq \min(n, Np),$$

$$E(X) = np,$$

$$\text{Var}(X) = np(1 - p)\left(1 - \frac{n-1}{N-1}\right).$$

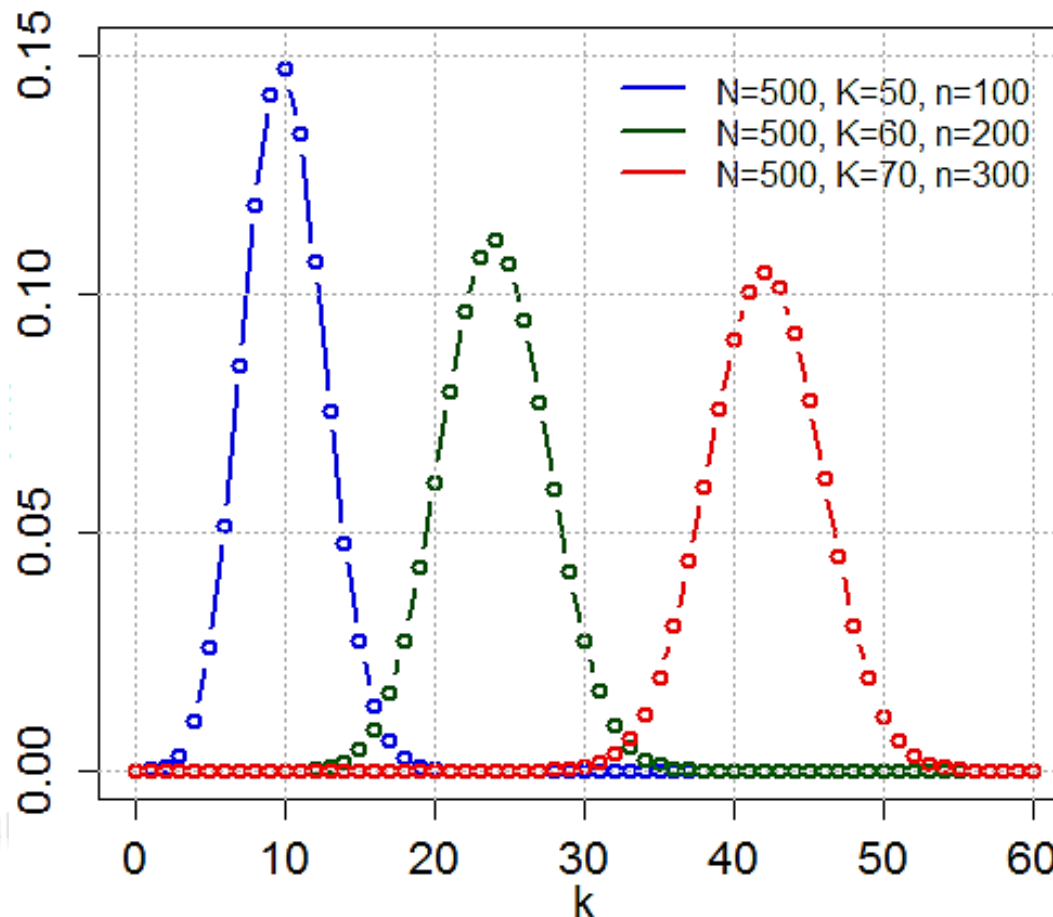
Random variables

- Hypergeometric distribution
 - Compared to the variance of the binomial distribution the last formula contains the factor $1 - \frac{n-1}{N-1}$ which is called the finite population correction
 - Despite the dependence between the drawings without replacement, there is a normal approximation for the hypergeometric distribution:

$$\text{Hg}(N, n, p) \approx N\left(np, \sqrt{np(1-p)\frac{N-n}{N-1}}\right), \text{ if both } np \geq 5 \text{ and } n(1-p) \geq 5$$

Random variables

- Hypergeometric distribution



Random variables

- Multinomial distribution

- Multinomial distribution $(X_1, \dots, X_r) \sim Mn(n; p_1, \dots, p_r)$ is an extension of the binomial distribution

$$P(X_1 = k_1, \dots, X_r = k_r) = \binom{n}{k_1, \dots, k_r} p_1^{k_1} \dots p_r^{k_r}. \quad E(X_i) = np_i.$$

- It corresponds to n independent trials with r possible outcomes
 - If each trial has distribution $(p_1, \dots, p_r)$, then X_i gives the number of trials with outcome labeled by i . So that

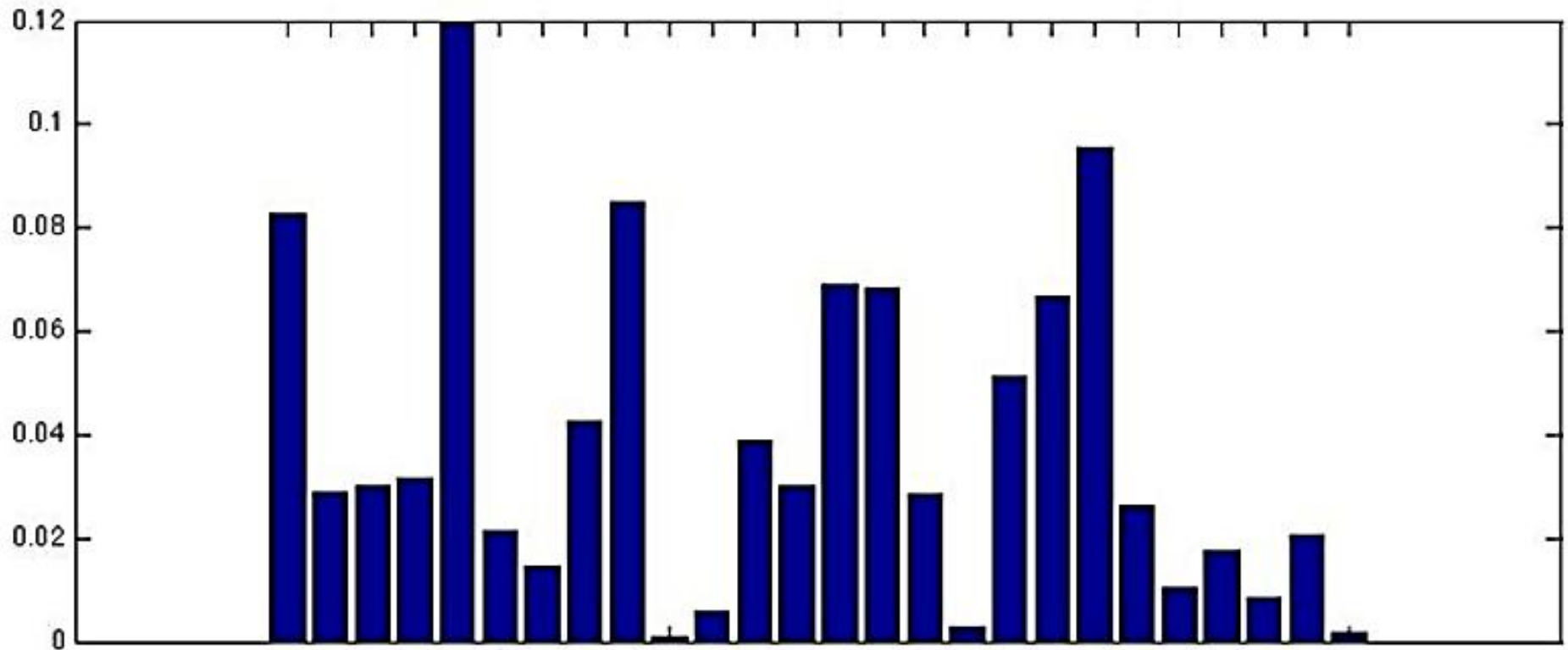
$$X_1 + \dots + X_r = n.$$

- The marginal distributions are binomial $X_i \sim Bin(n; p_i)$, and

$$Cov(X_i, X_j) = -np_i p_j, \quad i \neq j.$$

Random variables

- Multinomial distribution



Random variables

- chi-square distribution (also chi-squared or χ^2 -distribution)
 - distribution with k degrees of freedom is the distribution of a sum of the squares of k independent standard normal random variables.
 - The chi-square distribution is a special case of the gamma distribution and is one of the most widely used probability distributions in inferential statistics, notably in hypothesis testing and in construction of confidence intervals
 - This distribution is sometimes called the central chi-square distribution, a special case of the more general noncentral chi-square distribution

Random variables

- chi-square distribution (also chi-squared or χ^2 -distribution)

If Y_i have normal independent distributions with mean 0 and variance 1, then

$$\chi^2 \equiv \sum_{i=1}^r Y_i^2$$

is distributed as χ^2 with r degrees of freedom. This makes a χ^2 distribution a gamma distribution with $\theta \equiv 2$ and $\alpha \equiv r/2$, where r is the number of degrees of freedom.

More generally, if χ_i^2 are independently distributed according to a χ^2 distribution with $r_1, r_2, \dots, r_k$ degrees of freedom, then

$$\sum_{j=1}^k \chi_j^2$$

is distributed according to χ^2 with $r \equiv \sum_{j=1}^k r_j$ degrees of freedom.

Random variables

- chi-square distribution (also chi-squared or χ^2 -distribution)

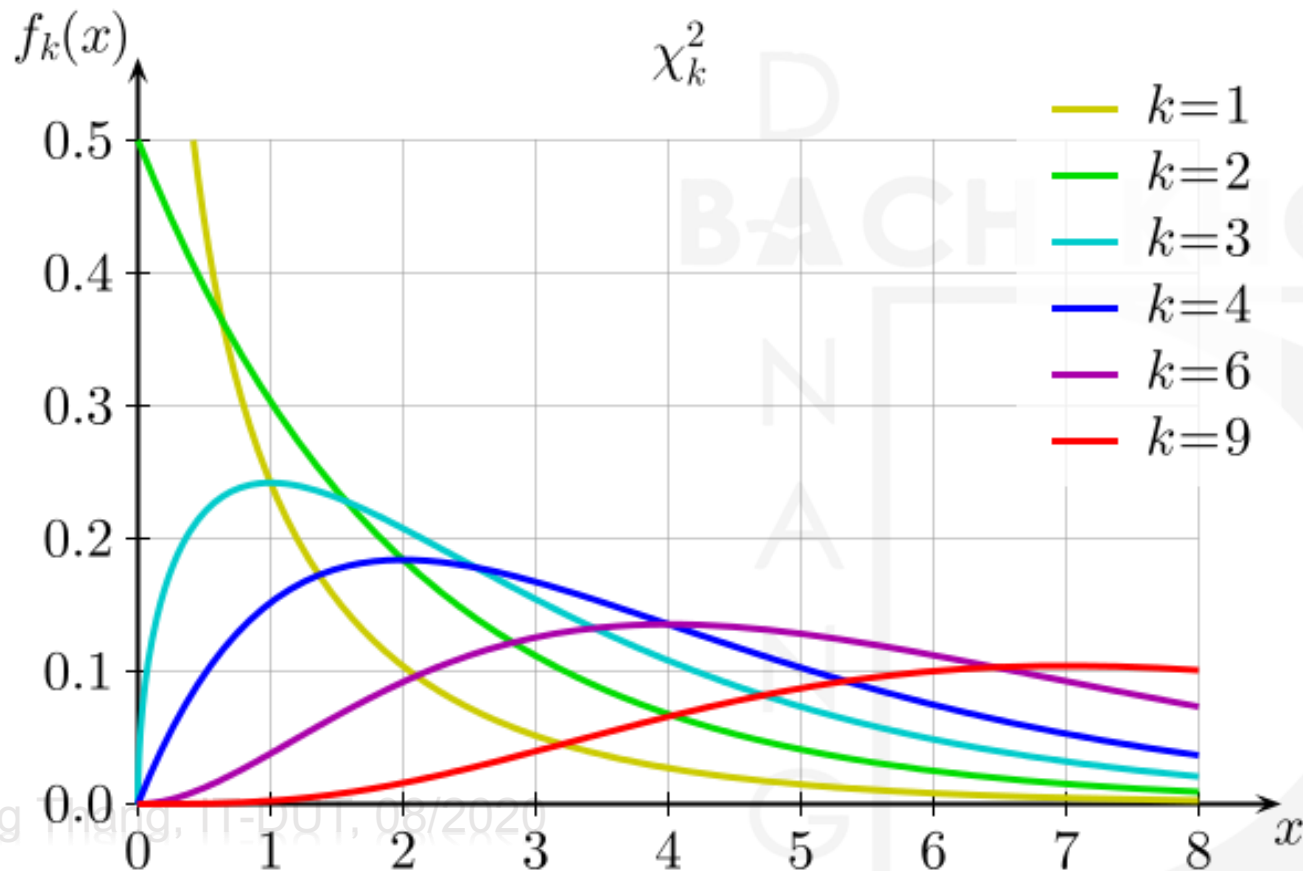
The probability density function (pdf) of the chi-square distribution is

$$f(x; k) = \begin{cases} \frac{x^{\frac{k}{2}-1} e^{-\frac{x}{2}}}{2^{\frac{k}{2}} \Gamma\left(\frac{k}{2}\right)}, & x > 0; \\ 0, & \text{otherwise.} \end{cases}$$

where $\Gamma(k/2)$ denotes the gamma function, which has closed-form values for integer k .

Random variables

- chi-square distribution (also chi-squared or χ^2 -distribution)



Random variables

- Gamma distribution
 - The gamma distribution is a two-parameter family of continuous probability distributions: The exponential distribution, Erlang distribution, and chi-squared distribution are special cases of the gamma distribution.
 - There are three different parametrizations in common use:
 - With a shape parameter k and a scale parameter θ .
 - With a shape parameter $\alpha = k$ and an inverse scale parameter $\beta = \frac{1}{\theta}$, called a rate parameter
 - With a shape parameter k and a mean parameter $\mu = k\theta = \frac{\alpha}{\beta}$
 - In each of these three forms, both parameters are positive real numbers.

Random variables

- Gamma distribution
 - The gamma distribution is the maximum entropy probability distribution (both with respect to a uniform base measure and with respect to a $\frac{1}{x}$ base measure) for a random variable X for which $E[X] = k\theta = \frac{\alpha}{\beta}$ is fixed and greater than zero, and $E[\ln(X)] = \psi(k) + \ln(\theta) = \psi(\alpha) - \ln(\beta)$ is fixed (ψ is the digamma function)

$$\psi(x) = \frac{d}{dx} \ln(\Gamma(x)) = \frac{\Gamma'(x)}{\Gamma(x)}$$

Random variables

- Gamma distribution
 - The gamma distribution can be parameterized in terms of a shape parameter $\alpha = k$ and an inverse scale parameter $\beta = \frac{1}{\theta}$, called a rate parameter
 - A random variable X that is gamma-distributed with shape α and rate β is denoted

$$X \sim \Gamma(\alpha, \beta) \equiv \text{Gamma}(\alpha, \beta)$$

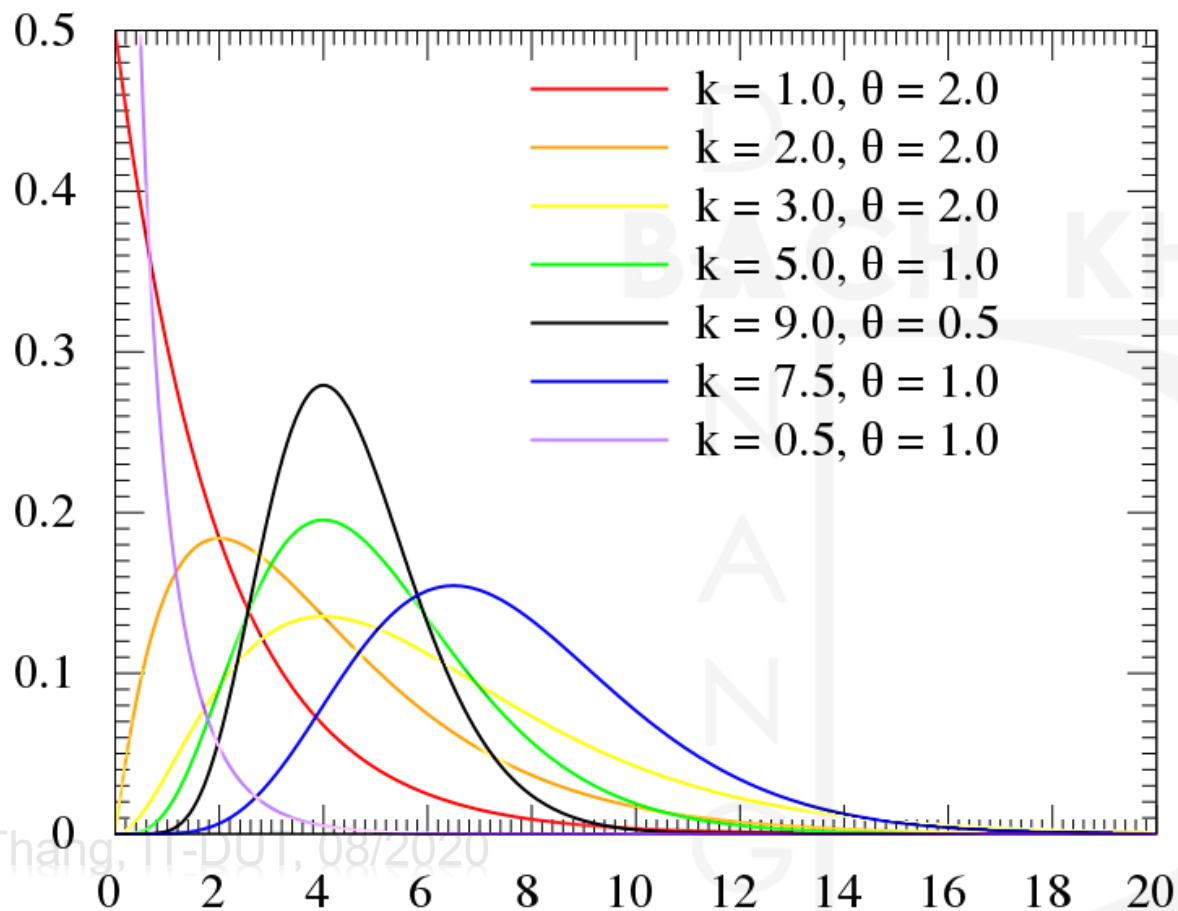
The corresponding probability density function in the shape-rate parametrization is

$$f(x; \alpha, \beta) = \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)} \quad \text{for } x > 0 \quad \alpha, \beta > 0,$$

where $\Gamma(\alpha)$ is the gamma function. For all positive integers, $\Gamma(\alpha) = (\alpha - 1)!$.

Random variables

- Gamma distribution



Probability

- Calculation
- Events
- Random variables
- **Markov chains**

Markov chains

- **A Markov chain:**

- A random process that consists of states and transitions between them. For each state, we know the probabilities for moving to other states
 - A process with a finite number of states (or outcomes, or events) in which the probability of being in a particular state at step $i + 1$ depends only on the state occupied at step i
 - *Prof. Andrei A. Markov* (1856 – 1922) , published his result in 1906.
- A Markov chain can be represented as a graph whose nodes are states and edges are transitions

Markov chains

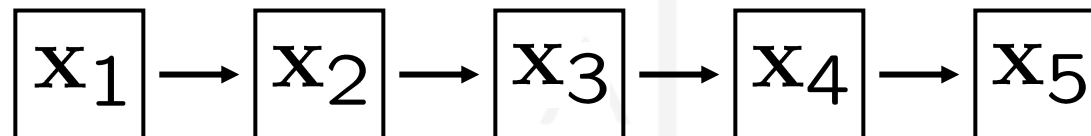
- **A Markov chain:**

- If the time parameter is discrete $\{t_1, t_2, t_3, \dots\}$, it is called Discrete Time Markov Chain (DTMC)
- If time parameter is continues, $(t \geq 0)$ it is called Continuous Time Markov Chain (CTMC)

Markov chains

- **A Markov chain:**

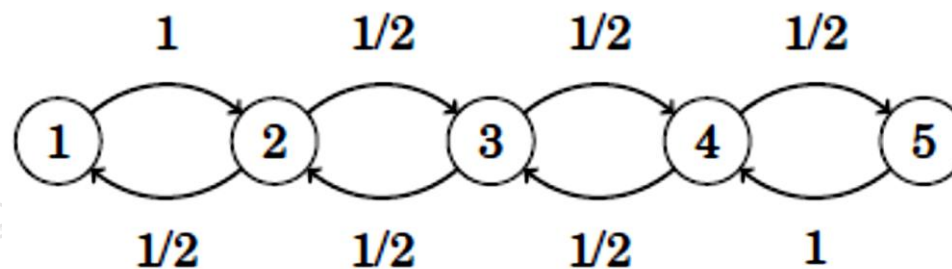
- A sequence of trials of an experiment is a Markov chain if:
 - The outcome of each experiment is one of a set of discrete states;
 - The outcome of an experiment depends only on the present state, and not on any past states.
 - A sequence of discrete random variables



- x_t is the **state** of the model at time t , each state is dependent only on the previous one

Markov chains

- A Markov chain
 - As an example, consider a problem where we are in floor 1 in an n floor building. At each step, we randomly walk either one floor up or one floor down, except that we always walk one floor up from floor 1 and one floor down from floor n
 - What is the probability of being in floor m after k steps?
 - In this problem, each floor of the building corresponds to a state in a Markov chain. For example, if $n = 5$, the graph is as follows:



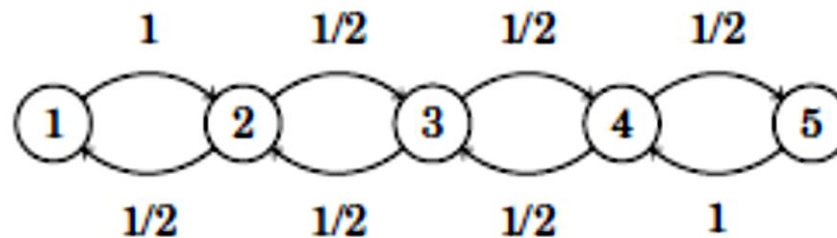
Markov chains

- A Markov chain
 - A Markov chain model is defined by
 - a set of **states**
 - We'll think of a traversal through the states as *generating* a sequence
 - Each state adds to the sequence (except for *begin* and *end*)
 - A set of **transitions** with associated probabilities
 - The transitions emanating from a given state define a distribution over the possible next states

Markov chains

- A Markov chain**

- The probability distribution of a Markov chain is a vector $[p_1, p_2, \dots, p_n]$, where p_k is the probability that the current state is k
 - The formula $p_1 + p_2 + \dots + p_n = 1$ always holds
 - In the above scenario, the initial distribution is $[1, 0, 0, 0, 0]$ because we always begin in floor 1.
 - The next distribution is $[0, 1, 0, 0, 0]$, because we can only move from floor 1 to floor 2. After this, we can either move one floor up or one floor down, so the next distribution is $[\frac{1}{2}, \frac{0, 1}{2}, 0, 0]$ and so on



Markov chains

- **A Markov chain**

- An efficient way to simulate the walk in a Markov chain is to use dynamic programming.
- The idea is to maintain the probability distribution, and at each step go through all possibilities how we can move. Using this method, we can simulate a walk of m steps in $O(n^2m)$ time

Markov chains

- **A Markov chain**

- The transitions of a Markov chain can also be represented as a matrix that updates the probability distribution
- **Transition Matrix** : contains all the conditional probabilities of the Markov chain

$$p_{ij} = \text{prob}(\text{state } n + 1 \text{ is } S_i \mid \text{State } n \text{ is } S_j)$$

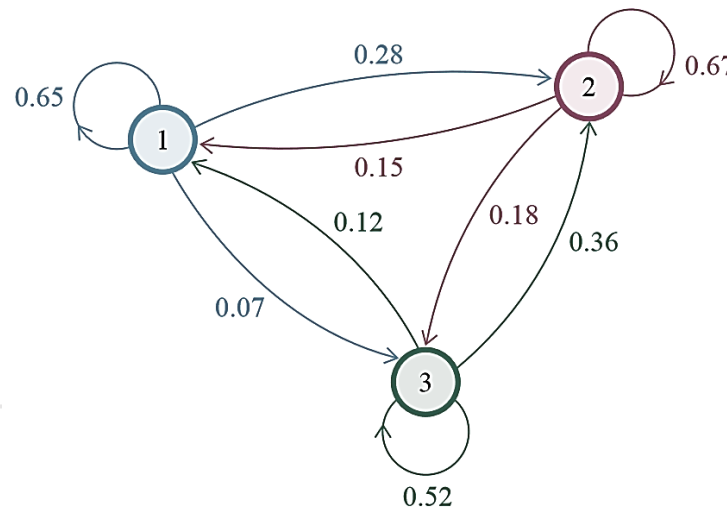
$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1r} \\ p_{21} & p_{22} & \cdots & p_{2r} \\ \vdots & \vdots & \ddots & \vdots \\ p_{r1} & p_{r2} & \cdots & p_{rr} \end{bmatrix}$$

where P_{ij} is the conditional probability of being in state S_i at step $n + 1$ given that the process was in state S_j at step n

Markov chains

- A Markov chain**

$$\text{State } n+1 \left\{ \begin{array}{c} S_1 \\ S_2 \\ S_3 \end{array} \right. \begin{array}{c} \overbrace{S_1 \quad S_2 \quad S_3}^{\text{State } n} \\ \begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{bmatrix} \end{array} \quad \begin{array}{c} 1 \\ 2 \\ 3 \end{array} \begin{bmatrix} 0.65 & 0.28 & 0.07 \\ 0.15 & 0.67 & 0.18 \\ 0.12 & 0.36 & 0.52 \end{bmatrix} = P.$$



Markov Chains

- **A Markov chain**

- **Transition matrix features**

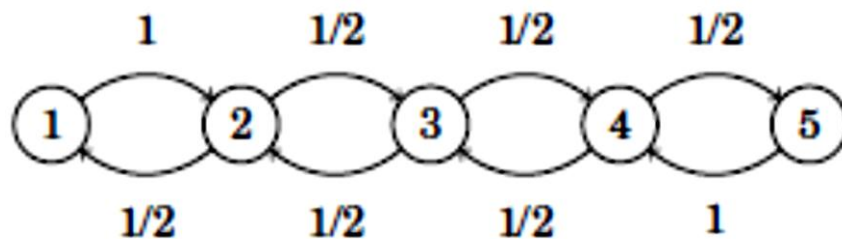
- It is square, since all possible states must be used both as rows and as columns
 - All entries are between 0 and 1, because all entries represent probabilities
 - The sum of the entries in any row must be 1, since the numbers in the row give the probability of changing from the state at the left to one of the states indicated across the top

Markov chains

- **A Markov chain**

- In the above scenario, the matrix is

$$\begin{bmatrix} 0 & 1/2 & 0 & 0 & 0 \\ 1 & 0 & 1/2 & 0 & 0 \\ 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1/2 & 0 & 1 \\ 0 & 0 & 0 & 1/2 & 0 \end{bmatrix}$$



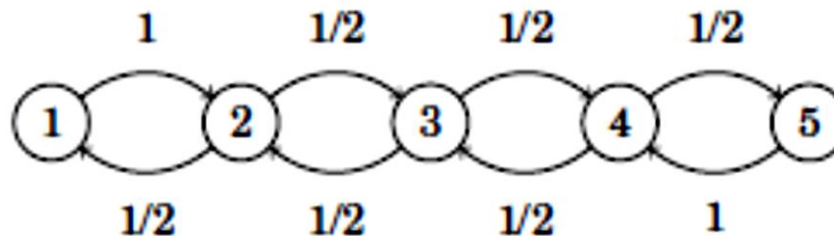
Markov chains

- A Markov chain**

- When we multiply a probability distribution by this matrix, we get the new distribution after moving one step. For example, we can move from the distribution $[1,0,0,0,0]$ to the distribution $[0,1,0,0,0]$ as follows:

$$\begin{bmatrix} 0 & 1/2 & 0 & 0 & 0 \\ 1 & 0 & 1/2 & 0 & 0 \\ 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1/2 & 0 & 1 \\ 0 & 0 & 0 & 1/2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

- By calculating matrix powers efficiently, we can calculate the distribution after m steps in $O(n^3 \log m)$ time



Markov chains

- **A Markov chain**

- Special cases of Markov chains:

- **Regular Markov chains:**

- A Markov chain is a regular Markov chain if some power of the transition matrix has only positive entries. That is, if we define the (i, j) entry of P^n to be p^n_{ij} , then the Markov chain is regular if there is some n such that $p^n_{ij} > 0$ for all (i, j)

- **Absorbing Markov chains:**

- A state S_k of a Markov chain is called an absorbing state if, once the Markov chains enters the state, it remains there forever

Markov chains

- **A Markov chain**

- Special cases of Markov chains:

- **A Markov chain is called an absorbing chain if**

- It has at least one absorbing state.

- For every state in the chain, the probability of reaching an absorbing state in a finite number of steps is nonzero

Markov chains

- A Markov chain**

- Special cases of Markov chains:

Regular

$$A = \begin{bmatrix} 0.75 & 0.25 & 0 \\ 0 & 0.5 & 0.5 \\ 0.6 & 0.4 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0.5625 & 0.3125 & 0.125 \\ 0.3 & 0.45 & 0.25 \\ 0.45 & 0.35 & 0.2 \end{bmatrix}$$

Since all entries in A^2 are positive, matrix A is regular.

Not regular

$$B = \begin{bmatrix} 0.5 & 0 & 0.5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

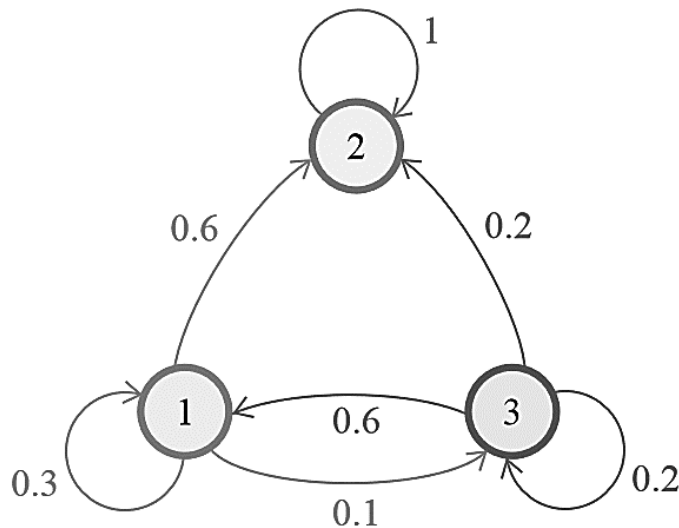
$$B^2 = \begin{bmatrix} 0.25 & 0 & 0.75 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; B^3 = \begin{bmatrix} 0.125 & 0 & 0.875 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; B^4 = \begin{bmatrix} 0.0625 & 0 & 0.9375 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Markov chains

- A Markov chain**

- Special cases of Markov chains:

State 2 is absorbing



$$p_{ii} = 2 \Rightarrow P_{22} = 1$$

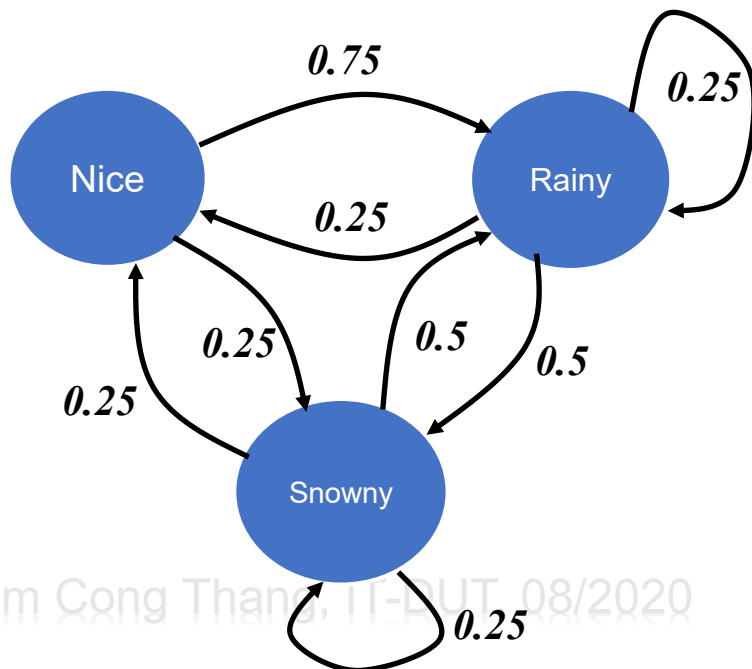
$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0.3 & 0.6 & 0.1 \\ 0 & 1 & 0 \\ 0.6 & 0.2 & 0.2 \end{bmatrix} \end{matrix} = P.$$

Markov chains

- Example: Springtime
 - Three possible conditions: nice, rainy, snowy
 - If it's nice today, then tomorrow it will be: rainy 75% of the time; snowy 25% of the time
 - If it's rainy today, then tomorrow it will be: rainy 25% of the time; nice 25% of the time; snowy 50% of the time
 - If it's snowy today, then tomorrow it will be: rainy 50% of the time; nice 25% of the time; snowy 25% of the time

Markov chains

- Example: Springtime
 - Three possible conditions: nice, rainy, snowy
 - We can represent this as a kind of graph



$$\begin{matrix}
 & \mathbf{x}_t \\
 \mathbf{x}_{t-1} & \begin{bmatrix} 0.0 & 0.75 & 0.25 \\ 0.25 & 0.25 & 0.5 \\ 0.25 & 0.5 & 0.25 \end{bmatrix}
 \end{matrix}$$

Transition probabilities

Markov chains

- Example: Springtime
 - Three possible conditions: nice, rainy, snowy
 - If it's nice today, what's the probability that it will be nice tomorrow?
 - If it's nice today, what's the probability that it will be nice the day after tomorrow?

$$\begin{matrix} & & & \mathbf{x}_t \\ & & & \downarrow \\ \mathbf{x}_{t-1} & & \begin{bmatrix} 0.0 & 0.75 & 0.25 \\ 0.25 & 0.25 & 0.5 \\ 0.25 & 0.5 & 0.25 \end{bmatrix} \end{matrix}$$

Transition probabilities

Markov chains

- Example: Springtime
 - Three possible conditions: nice, rainy, snowy

$$P = \begin{matrix} & \begin{matrix} N & R & S \end{matrix} \\ \begin{matrix} N \\ R \\ S \end{matrix} & \begin{bmatrix} 0.0 & 0.75 & 0.25 \\ 0.25 & 0.25 & 0.5 \\ 0.25 & 0.5 & 0.25 \end{bmatrix} \end{matrix}$$

- The transition matrix at time $t + 2$ is P^2

$$\begin{bmatrix} 0.0 & 0.75 & 0.25 \\ 0.25 & 0.25 & 0.5 \\ 0.25 & 0.5 & 0.25 \end{bmatrix} \begin{bmatrix} 0.0 & 0.75 & 0.25 \\ 0.25 & 0.25 & 0.5 \\ 0.25 & 0.5 & 0.25 \end{bmatrix} = \begin{bmatrix} 0.2500 & 0.3125 & 0.4375 \\ 0.1875 & 0.5000 & 0.3125 \\ 0.1875 & 0.4375 & 0.3750 \end{bmatrix}$$

- The transition matrix at time $t+n$ is P^n

Markov chains

- Example: Springtime
 - Three possible conditions: nice, rainy, snowy
 - the weather be like in 20 days?

$$P^{20} = \begin{bmatrix} 0.2 & 0.44 & 0.36 \\ 0.2 & 0.44 & 0.36 \\ 0.2 & 0.44 & 0.36 \end{bmatrix}$$

- Almost completely independent of the weather today
- The row [0.2 0.44 0.36] is called the stationary distribution of the Markov chain

Markov chains

- Markov Chain Models

- Let X be a sequence of L random variables $X_1 \dots X_L$ representing a biological sequence generated through some process
- How probable the sequence is given our model or any probabilistic model of sequences, we can write this probability as (recall the “Chain Rule of Probability”)?

$$\begin{aligned}\Pr(X) &= \Pr(X_L, X_{L-1}, \dots, X_1) \\ &= \Pr(X_L | X_{L-1}, \dots, X_1) \Pr(X_{L-1} | X_{L-2}, \dots, X_1) \dots \Pr(X_1)\end{aligned}$$

- key property of a (1st order) Markov chain: the probability of each X_i depends only on the value of X_{i-1}

$$\begin{aligned}\Pr(X) &= \Pr(X_L | X_{L-1}) \Pr(X_{L-1} | X_{L-2}) \dots \Pr(X_2 | X_1) \Pr(X_1) \\ &= \Pr(X_1) \prod_{i=2}^L \Pr(X_i | X_{i-1})\end{aligned}$$

Markov chains

- Markov Chain Notation

- The transition parameters can be denoted by $a_{x_{i-1}x_i}$ where

$$a_{x_{i-1}x_i} = \Pr(X_i = x_i | X_{i-1} = x_{i-1})$$

- Similarly we can denote the probability of a sequence x as

$$a_{Bx_1} \prod_{i=2}^L a_{x_{i-1}x_i} = \Pr(x_1) \prod_{i=2}^L \Pr(x_i | x_{i-1})$$

where a_{Bx_1} represents the transition from the *begin* state

- This gives a probability distribution over sequences of length L

Markov chains

- Estimating the Model Parameters
 - Given some data, how can we determine the probability parameters of our model?
 - One approach: maximum likelihood estimation
 - given a set of data D
 - set the parameters θ to maximize

$$\Pr(D|\theta)$$

i.e. make the data D look as likely as possible under the model

Markov chains

- Maximum likelihood estimation
 - Let's use a very simple sequence model
 - every position is independent of the others
 - every position generated from the same multinomial distribution
 - We want to estimate the parameters $\Pr(a), \Pr(c), \Pr(g), \Pr(t)$ and we're given the sequences **accgcgcttagcttagtgactagccgttac**

$$\Pr(a) = \frac{n_a}{\sum_i n_i}$$

then the maximum likelihood estimates are the observed frequencies of the bases

$$\Pr(a) = \frac{6}{30} = 0.2$$

$$\Pr(g) = \frac{7}{30} = 0.233$$

$$\Pr(c) = \frac{9}{30} = 0.3$$

$$\Pr(t) = \frac{8}{30} = 0.267$$

Markov chains

- Maximum likelihood estimation
 - Suppose instead we saw the following sequences

gccgcgcttggcttgggtggctggccgttgc

- Then the maximum likelihood estimates are

$$\Pr(a) = \frac{0}{30} = 0$$

$$\Pr(c) = \frac{9}{30} = 0.3$$

$$\Pr(g) = \frac{13}{30} = 0.433$$

$$\Pr(t) = \frac{8}{30} = 0.267$$

do we really want to set this to $\Pr(a) = 0$?

Markov chains

- Maximum likelihood estimation: A Bayesian Approach
 - Instead of estimating parameters strictly from the data, we could start with some prior belief for each
 - for example, we could use *Laplace estimates*

$$\Pr(a) = \frac{n_a + 1}{\sum_i (n_i + 1)}$$

Where n_i represents the number of occurrences of character i

- Using Laplace estimates with the sequences

$$\Pr(a) = \frac{0 + 1}{34}$$

$$\Pr(c) = \frac{9 + 1}{34}$$

Markov chains

- Maximum likelihood estimation: A Bayesian Approach

$$\Pr(a) = \frac{n_a + p_a m}{\left(\sum_i n_i \right) + m}$$

p_a : prior probability of a
 m : number of “virtual” instances

- with $m=8$ and uniform priors

gccgcgcttggcttggctggcggttgc

$$\Pr(c) = \frac{9 + 0.25 \times 8}{30 + 8} = \frac{11}{38}$$

Markov chains

- Maximum likelihood estimation: A Bayesian Approach
 - Estimation for 1st Order Probabilities
 - to estimate a 1st order parameter, such as $\Pr(c|g)$, we count the number of times that c follows the history g in our given sequences
 - using Laplace estimates with the sequences

gccgcgcttggcttgggtggctggccgttgc

$$\Pr(a | g) = \frac{0 + 1}{12 + 4}$$

$$\Pr(g | g) = \frac{3 + 1}{12 + 4}$$

$$\Pr(a | c) = \frac{0 + 1}{7 + 4}$$

⋮

$$\Pr(c | g) = \frac{7 + 1}{12 + 4}$$

$$\Pr(t | g) = \frac{2 + 1}{12 + 4}$$